

BCSC1002: OBJECT ORIENTED PROGRAMMING

Objective: This course introduces the Object-Oriented programming paradigm to students. It also teaches a student how to think objectively and model a Java program for solving real-world problems.

Prerequisites: Basic programming, control structures, functions, and a suitable language like Python or Java.

CREDITS:03

L-T-P-J:3-0-0-0

Mod ule No	Content	Teachi ng Hour s
I	Object-Oriented Programming: Features of Object-Oriented Programming, Introduction to Object-Oriented Java Programming. Java Technology & Environment: Understanding the compilation process of the JVM, JVM vs JDK vs JRE, Key Features of Java, Structure of a simple Java program. Working with Java Primitive Data Types: Strongly Typed nature of Java, Primitive Data Types in Java, The new 'var' keyword, Scope of a variable. Accepting User Input in Java Programs: using the Scanner class, using command line arguments. Programming Constructs: Sequence, Selection, Iteration & Transfer Statements, For-Each Loop. Working with Java Arrays: Declaring and Initializing One-Dimensional and Two-Dimensional Arrays in Java, Introduction to java. util. Arrays class. The String API: String Data Type, commonly used methods from the String API, String Tokenizer, String Builder & String Buffer. Creating and Using Methods: Signature of a method, Types of Methods, Overloading methods in a class, Static and Non-Static Methods. Describing and Using Objects & Classes: Declare the structure of a Java class, declaring members of a class (fields and methods), declaring and using Java Objects, lifecycle of an Object (creation, assignment, dereferencing and garbage collection), Constructors of a class, Overloading Constructors, Constructor chaining using 'this' and 'super' keyword. Using Java Packages: create and import Java packages and static imports, abstracting program logic to packages, creating executable main class, running the executable class inside a package. Applying Encapsulation: Using access modifiers with/in a class, principles of encapsulation. Programming Abstractly Through Interfaces: create and implement Interfaces for programs, private and default methods in Interfaces, declaring Abstract Classes, Constructors in Abstract Classes. Marker Interface, Functional Interfaces, Lambda Expressions in Java.	20

II	Reusing Implementations using Inheritance: Declaring Subclasses and Super classes, extend Abstract Classes, implementing Interfaces, exploring polymorphic behavior by overriding methods, Object Types vs Reference Types, differentiate overloading, overriding and hiding. Exception Handling: Exception Hierarchy, Need of Exception Handling, Checked Exceptions, Unchecked Exceptions and Errors, Try-Catch Blocks, Finally, Throw & Throws Keywords, creating and handling Custom Exceptions. Threads in Java: Life Cycle of a Thread, creating threads using Runnable and Thread, 'sleep ()', Thread Priorities. Using Wrapper Classes: Wrapper Classes in Java, Boxing-Unboxing-Auto Boxing-Auto Unboxing. Generics & Collections: Creating Generic classes, Generic Methods, Diamond Notation, Wildcards, Type Erasure, Collection Hierarchy, Base Interfaces, Lists, Sets and Maps. The Stream API: Introduction to the Stream API, using lambda expressions in Streams. Regular Expressions: Pattern and Matcher Class. JDBC: JDBC Drivers, Connecting to a MySQL Database, Driver Manager, Connection Interface, Statement Interface, Result Set Interface, Prepared Statements.	20
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TEXT BOOK

- Herbert Schildt, "The Complete Reference, Java Eleventh Edition", Oracle Press.2019.

Reference Book:

- Cay S Hosrtnmann, "Core Java Volume I—Fundamentals, Eleventh Edition", Pearson,2018.
- Rogers Cadenhead, "Sams Teach Yourself Java in 21 Days (Covers Java 11/12), 8th Edition", Pearson,2020.

Focus: This Course focus on Employability under all COs.

Course Outcomes: After completion of the course, students will be able to -

- CO1: Understand the basics of Object-Oriented Programming paradigm.
- CO2: Construct the logical flow of programs by using the sequence, selection, iterations and transfer statements.
- CO3: Apply the concepts of Object-Oriented Programming to model programs in Classes, Abstract Classes, Interfaces and Enums, and simplify program function by dissecting it into methods.
- CO4: Understand accessibility of members in a program unit and create packages to prevent namespace collisions.
- CO5: Predict run-time errors in a program by examining program functioning.
- CO6: Show the parallel processing capabilities of a program using a multithreading concept.
- CO7: Experiment with the predefined classes and interfaces defined in the Collections Framework.
- CO8: Develop a program using JDBC connectivity to demonstrate data persistence.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

Cos	POs/PSOs
CO1	PO1,PO3/PSO1,PSO2
CO2	PO1,PO3/PSO1,PSO2
CO3	PO1,PO2/PSO1,PSO2
CO4	PO1/PSO2,PSO4
CO5	PO1,PO2,PO4/PSO4
CO6	PO1,PO2, PO3/PSO2
CO7	PO1,PO2,PO11/PSO2
CO8	PO1,PO2,PO3/PSO1,PSO2

BCSC0801: OBJECT ORIENTED PROGRAMMING LAB

Objective: The objective of this course is that students will study and learn Object Oriented Modeling and programming.

Credits:01

L-T-P-J:0-0-2-0

Mod ule No	Content	Teaching Hours
I & II	Programs in Java and python based on the concepts of: - Classes, Constructors, Polymorphism and Keyword Static. Programs based on the concepts of: - Inheritance, Multithreading Using Thread Class & Interface Runnable, String Handling, Generic Classes. Programs based on the concepts of: - Handling Database Connectivity. - Implementation of Collection Framework. Programs based on the concepts of: - Database Connectivity. - Retrieving Data from Database. - Parameters Passing, Execute many Method. - Cursor Attributes. - Invoke Stored Procedures. - Invoke Stored Functions.	24

Reference Books:

- Naughton, Schildt, "The Complete Reference JAVA2", 9th Edition, Oracle Press.
- Bhave&Patekar, "Programming with Java", Pearson Education
- Bret Slatkin: "Effective Python: 59 Specific ways to write better Python", Addison Wesley, 2015.

Focus: This Course focus on Employability under all COs.

Course Outcome: After completion of course, the student will be able to:

- CO1: Implement object oriented language features.
- CO2: Design GUIs and Graphical programming.
- CO3: Design object oriented solutions for small systems involving database and event handling concepts.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

Cos	POs/PSOs
CO1	PO1,PO2/PSO1
CO2	PO3,PO5/PSO2
CO3	PO3,PO5/PSO4